

Tutorial of binding affinity prediction by MolAICal and Pafnucy model

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1. Introduction

Deep learning model can be used to evaluate binding affinity between ligands and receptors. In this tutorial, Pafnucy [1] is introduced to calculate binding affinity of ligands by MolAICal. This example can be for users to know quickly how deep learning predicts binding affinity between ligands and proteins. This tutorial can only run on Linux system. For more detailed MolAICal, please visit: <https://molaical.github.io>.

2. Materials

2.1. Software requirement

- 1) MolAICal: <https://molaical.github.io>
- 2) UCSF Chimera: <https://www.cgl.ucsf.edu/chimera/>
- 3) VMD: <https://www.ks.uiuc.edu/Research/vmd/>

2.2. Example files

- 1) All the necessary tutorial files are downloaded from:
<https://github.com/MolAICal/tutorials/tree/master/015-bindingaffinityPafnucy>

3. Procedure

3.1 Install Pafnucy model

Currently, only Linux version of MolAICal is supported.

- 1) Click and open website: **DownloadModel**
Then go to the folder “[AImodels--BindingAffinity--pafnucy--linux](#)” in the open website, download a file named “pafnucy.tar.gz”
- 2) Move “pafnucy.tar.gz” to the folder “MolAICal-xxx/mtools” in the MolAICal container.
Where “MolAICal-xxx” is your decompressed root directory of MolAICal in the MolAICal container. The folder “mtools” is the targeted directory. The movement procedures are as follows:

a) First of all, copy the files to the MolAICal container.

```
***** 1. Go to the container file system, it will enter the "/root" *****
#> molaical.exe -eset shell in

***** 2. The first part of 'cp' is from the local machine, and the second part is in the container *****
#> mv <Your machine path>/pafnucy.tar.gz /root/soft/MolAICal-xxx/mtools

***** 3. Exit container file system *****
#> exit
```

b) Secondly, enter the container virtual environment of MolAICal:

```
#> molaical.exe -eset sys run molaical  
#> cd /root/soft/MolAICal-xxx/mtools
```

Note: 'molaical' is the container name.

3) Decompress file

```
#> tar -xzf pafnucy.tar.gz
```

And go to folder named Pafnucy:

```
#> cd pafnucy
```

4) install Pafnucy model

```
#> chmod +x install.sh
```

```
#> ./install.sh
```

Until now, Pafnucy model is installed completely.

✧ Remember to use the "exit" command to quit the MolAICal virtual environment and return to the local computer for computation (primarily to avoid file-copying steps; **execution within the container is also possible but requires copying data from the local computer to the MolAICal container**).

```
#> exit
```

3.2 Calculate binding affinity for one complex

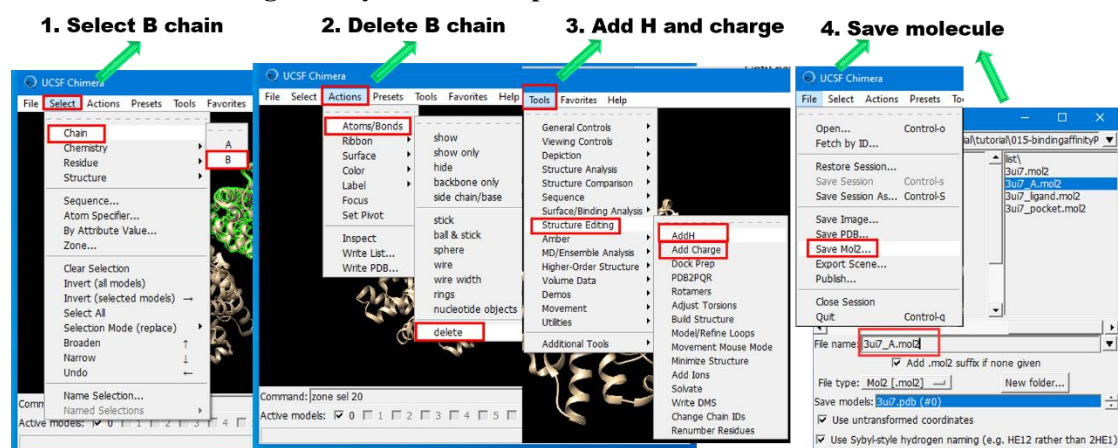


Figure 1. Deal with protein file

3.2.1 Prepare protein file

1) "3ui7.pdb" is a protein file that is downloaded from PDB database. First of all, open UCSF Chimera and save A chain of "3ui7.pdb" as "3ui7_A.mol2" (see Figure 1).

(Notice: Here, assuming **hydrogens** and **charge** have been added, where **Amber ff14SB** is for standard residues and **AM1-BCC** is for non-standard residues and ligands in UCSF Chimera

(see Figure 1).)

2) Get the geometric center of ligand (see Figure 2). Geometric center is 5.411, 12.284, 43.559

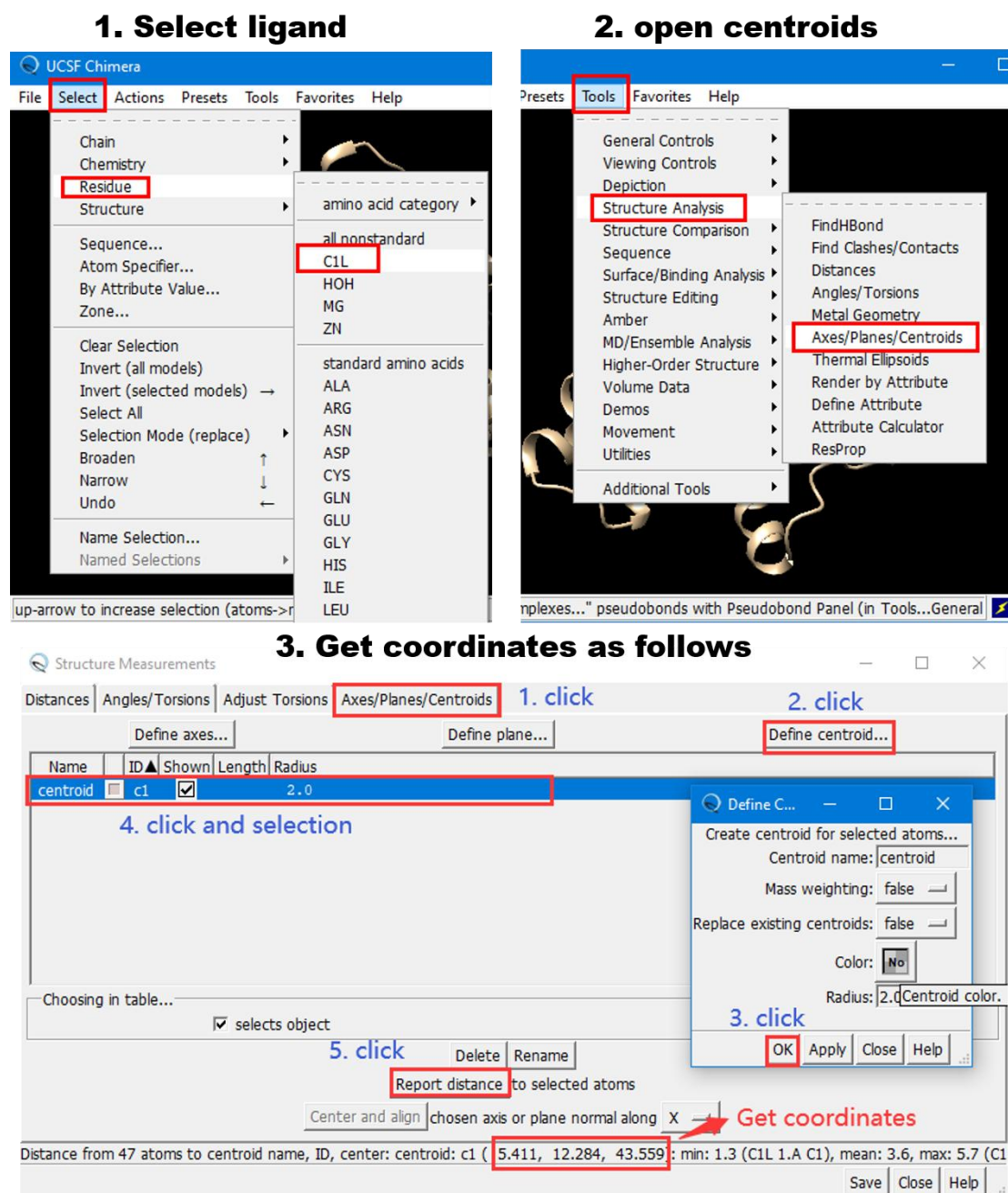


Figure 2

3) The original paper of Pafnucy [1] cropped the complex to a defined size of 20 Å cubic box around the geometric center of a ligand. Get maximum and minimum coordinates of 20 Å cubic box via MolAICal (Notice: the double quotes are necessary for X, Y, Z coordinates. The interval distance among X, Y, Z coordinates and box lengths should be one space):

```
#> molaical.exe -tool box -i "5.411 12.284 43.559" -l "20.0 20.0 20.0" -o box.bild
```

It will generate a file named “box.bild” which contains minimum and maximum coordinates in

“.box” part. Open “box.bild”, it can be find like this in “.box” part: -4.589 2.2840000000000007 33.559 15.411 22.284 53.559

4) load file “3ui7_A.mol2” and open Tk Console of VMD (see Figure 3).

1. New molecule 2. Browse protein and load it

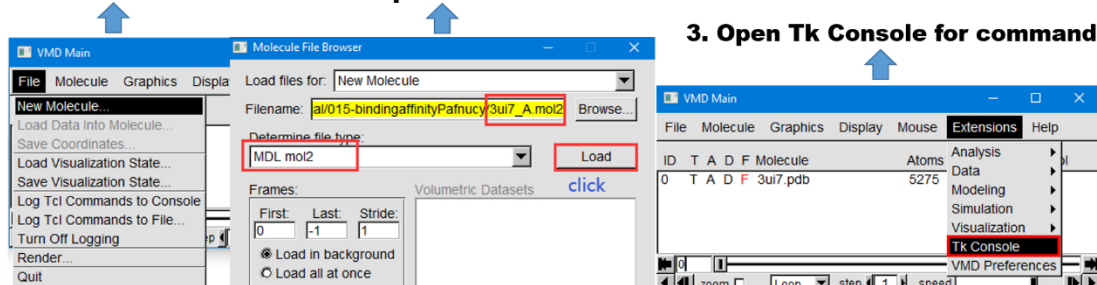


Figure 3

5) Using VMD to save the complex that is within 20 Å cubic box around the geometric center of a ligand.

My workspace is in E: disk, so I use below commands to change my workspace in VMD TkConsole:

```
#> cd E:
```

```
#> cd workdir/MolAICal/tutorial/tutorial/015-bindingaffinityPafnucy/
```

Input command below in VMD TkConsole (**Note:** minimum and maximum x, y, z of box have been calculated above, users can replace their own real data in the below command):

```
#> set box [atomselect top "(not ((x < -4.589 or x > 15.411) or (y < 2.2840000000000007 or y > 22.284) or (z < 33.559 or z > 53.559)))"]
```

```
#> $box writemol2 3ui7_pocket.mol2
```

Figure 4 shows records in VMD Tk Console for this tutorial. Save selected molecule to “3ui7_pocket.mol2”. Please check it. Users should use their own minimum and maximum data.

(Discussion: Sometimes, I see somebodies employ full protein in mol2 for binding affinity prediction using Pafnucy model. If yes, users can omit above five steps.)

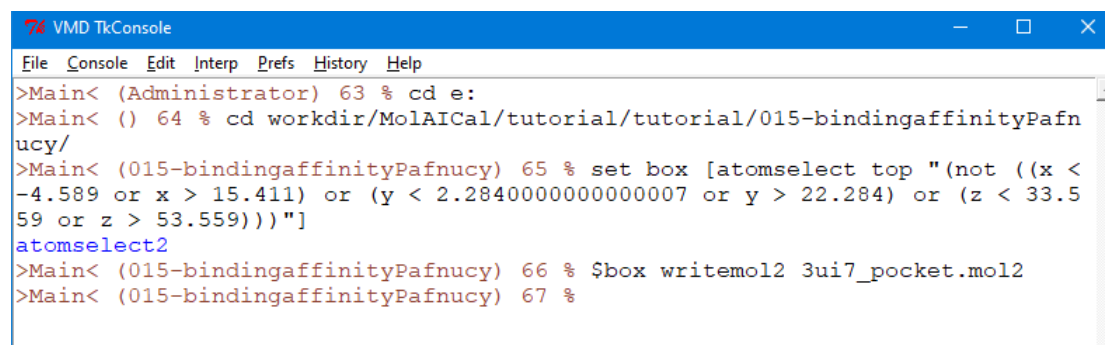


Figure 4

3.2.2 Calculate binding affinity of one protein and one ligand

Go to [015-bindingaffinityPafnucy](#), and use below command:

```
#> molaical.exe -model pafnucy -l 3ui7_ligand.mol2 -p 3ui7_pocket.mol2 -o results.csv
```

It will generate a file named results.csv that contains pK_x (pK_d or pK_i) value of one complex. If users want to convert pK_x to binding free energy, they can refer to “<https://molaical.github.io/tutorial.html>” and MolAICal manual which part is involved in “Binding free energy from pK_d or pK_i”.

Notice: if users do not have mol2 format molecular file for ligands, they use the command of MolAICal to convert molecules in mol2 format. For instance, users have pdb format file, they can use the below command for format conversion (**Note:** molecule should have correct suffix so that MolAICal can recognize them automatically):

```
#> molaical.exe -tool format -i ligand.pdb -o ligand.mol2
```

3.2.3 Calculate binding affinity of many ligands and one protein

Sometimes, users want to calculate binding affinities of many ligands, under these circumstances, it can employ Linux shell to realize it easily.

Go to [015-bindingaffinityPafnucy/list](#), and input below command:

```
#> ls lig*.mol2 > list.txt
```

It will generate a file named “list.txt” which contains ligand names. Then, [open the file named “run.sh”](#), and modify right path of molaical.exe (see Figure 5)

```
#!/bin/bash
i=0
cat list.txt | while read line
do
    echo $line
    /bai/soft/molaicalv11/molaical.exe -model pafnucy -l $line -p pocket.mol2 -o $line'.csv'
    let i+=1
done
```

Figure 5

Finally run command as below:

```
#> bash run.sh
```

Merge all results using below command:

```
#> cat *.csv > results.csv
```

```
#> sed -i -e '/name,prediction/d' results.csv
```

```
#> sed -i '1 i\name,prediction' results.csv
```

It will generate a file named “results.csv” that contains the predicted pK_x values. If users want to convert pK_x to binding free energy, they can refer to “<https://molaical.github.io/tutorial.html>” and MolAICal manual which part is involved in “Binding free energy from pK_d or pK_i”.

3.2.4 Calculate binding affinity of few ligands and one protein in command line

If users do not have huge of ligands, they can use one command to calculate binding affinity between few ligands and one protein in the command line:

Go to [015-bindingaffinityPafnucy/list](#), and input the below command:

```
#> molaical.exe -model pafnucy -l "lig_1.mol2 lig_2.mol2" -p receptor.mol2 -o twoTest.csv
```

It will generate “twoTest.csv” which contains two binding affinity values of ligand files called lig_1.mol2 and lig_2.mol2. Please note that “[lig_1.mol2 lig_2.mol2](#)” should have double quotation marks in the command line. For more detailed commands, please check MolAICal manual.

Appendix

The linux shell content of “[run.sh](#)”:

```
#!/bin/bash

i=0
cat list.txt | while read line
do
    echo $line
    ~/bai/soft/molaicalv11/molaical.exe -model pafnucy -l $line -p pocket.mol2 -o $line'.csv'
    let i+=1
done
```

Reference

1. Stepniewska-Dziubinska MM, Zielenkiewicz P, Siedlecki P. Development and evaluation of a deep learning model for protein-ligand binding affinity prediction. *Bioinformatics*. 2018;34(21):3666-74.